

SECTION 15060-A
PIPE AND PIPE FITTINGS - GENERAL

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the requirements necessary to furnish and install the plant piping systems specified in Contract. The work shall include all pipe, tubing, hangers, supports, restraints, isolators, pipe pre-cleaning, testing, and final cleaning. In addition, this Section covers the installation of in-line valves and instruments and piping specialties.
- B. Line Class Designations: Table 1.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for the project piping systems.
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.

1.3 QUALITY ASSURANCE

- A. Manufacturer shall maintain a program to monitor the manufacturing process, in-process product quality, and final product quality. The program shall have the following features:
 - 1. Change control process.
 - 2. Standard manufacturing procedures.
 - 3. Standard calibration and test methods.
 - 4. Calibration records.
 - 5. Calibration standard traceability.
 - 6. A sampling plan.
 - 7. Test records.
 - 8. Documentation of upset conditions and corrective actions.
- B. For components with additional cleaning and packaging and/or testing requirements, monitor the following as appropriate:
 - 1. Quality of cleaning and packaging materials.
 - 2. Cleanliness of the cleaning and packaging facilities.

3. Surface finish of component bodies and tube stubs in Ra.
4. Out-gassing and/or leaching rates.
5. Particle generation.
6. Chrome-to-iron ratio, chrome-oxide to iron-oxide ratio, and chromium enrichment depth.
7. Surface flaw counts by SEM.
8. Surface contaminants.
9. Porosity by SEM.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Components with Special Cleaning/Packaging Requirements:
 1. Handle and store material and equipment in a controlled environment throughout the progress of the job to prevent damage and/or contamination.
 2. Repackage incoming material, which is defective or contaminated, document the problem, and provide such documentation to the Engineer and quality assurance representative (QAR) for final disposition.
 3. Do not use material or equipment that becomes damaged or contaminated in handling on the job unless it is repaired or re-cleaned to the original purchase requirements at Contractor expense. Segregate such material from the new, clean material for re-inspection and approval by the QAR before use.

1.5 SUBMITTALS

- A. Provide bolt torque tables for each gasket type by size and service per general conditions and before any gaskets are installed.
- B. When special cleaning and /or testing procedures are specified, provide as appropriate:
 1. Quality specifications for DI water, nitrogen, argon, helium, IPA, oil-free air, and other materials used in the cleaning or testing process.
 2. Description of the manufacturer's cleaning procedures, testing procedures, sample plan, and quality assurance program.
 3. Specifications for the cleanroom used to prepare, test, and package the components.
 4. Detailed description of the procedures and materials used in packaging.
 5. Leak rate (external and across the seat) test method and guarantees.
 6. Interior surface finish, roughness, and chrome-oxide levels test method and guarantees.
 7. Out-gassing test methods and guarantees for H₂O and hydrocarbons.

8. Particle performance test methods and guarantees.
9. Provide the following items with the shipment as appropriate:
 - a. Certificate of compliance.
 - b. Alloy, heat number, and sulfur content of tube extensions.
 - c. Leak test results.
10. Confirmation the component was assembled with no lubricants.

PART 2 - PRODUCTS

2.1 PIPE CLASS

- A. The pipe class specification sheets provided at the end of each detail piping system specification identify the pipe, fittings, and miscellaneous materials required for each pipe class. A two-letter/two-number code is used. The first two letters identify pipe material; the two numbers designate unique pipe data sheet identifiers (see Table 1 at the end of this Section).

2.2 CLEANING/PACKAGING LEVELS FOR HIGH-PURITY SERVICE

- A. Specific cleaning and packaging criteria apply only to valves and components that specifically reference the criteria in the valve or component data sheet or schedule. The criteria apply to valve and component manufacturing requirements prior to shipment.
- B. Level CP-1:
 1. Factory Cleaning Procedures - Prior to final assembly, clean parts as follows:
 - a. Hot DI flowing rinse where the DI water is greater than 18 meg ohm-cm before heating to 80 degrees C (26 degrees C). DI water quality criteria are specified in Part 3 of this Section. Flow rate shall be sufficient to remove all residual solvent traces.
 - b. Dry with 99.999 percent pure nitrogen, filtered with 0.01-micron-rated filters until moisture is 3 ppm or less.
 - c. Purge with HP cryogenic nitrogen or argon (99.999 percent minimum purity, filtered to 0.01-micron absolute). Double bag in sealed polyethylene bags (10-mil-thick inner bag) which have been purged with HP inert gas. Where sharp edges exist on components, additional padding shall be provided to prevent puncture of bags in shipping.

2. Final Packaging:
 - a. Following the DI cleaning process, unpackaged components shall only be exposed to HEPA filtered, Class 10 laminar flow regions. Components may only be removed from the Class 10 environment in sealed, non-shedding containers (work-in-progress containers or final shipment bagging).
 - b. Perform packaging in the same cleanroom where the cleaning process was completed. Do not remove components and fittings from the cleanroom after cleaning until they are properly packaged and sealed.
- C. Level CP-2:
1. Perform cleaning in a segregated Class 1,000 (M4.5) environment, away from manufacturing activities that generate dust or oil spray. Proper cleanroom attire and protocol shall be maintained at all times. Cover the work surface with clean polyethylene sheeting and replace whenever it becomes contaminated. Remove visible traces of contaminants using the cleaning procedure.
 2. Electronics-grade isopropyl alcohol (IPA) or DI water is the only acceptable cleaning fluids. Wipes shall be lint-free, cleanroom grade. Blow components dry using 0.2-micron filtered nitrogen. Do not use compressed air.
 3. Prior to packaging, inspect 100 percent of the components for contamination, manufacturing defects, and impurities. If a defect or contaminant is discovered, reject the component for high purity service.
 4. Components shall be completely dry before placement in double 4-mil polyethylene or Nylon 6 bags. Individually heat seal bags shut with the inner bag partially inflated with 0.2-micron-filtered nitrogen. Affix a label on each bag stating the component has been assembled without lubricants and cleaned for microelectronics service.
- D. Level CP-3:
1. Clean, assemble, test, and package in a Class-1,000 (M4.5) environment at a minimum.
 2. Use of lubricants for component or O-ring assembly is prohibited.
 3. Locate cleanroom away from dust- or oil-generating activities and comply with the following:
 - a. Maintain work surfaces clean and oil-free.

- b. DI water, electronics-grade isopropyl alcohol (IPA), and filtered oil-free air or nitrogen is the only acceptable cleaning fluids.
 - c. Wipes shall be lint-free and cleanroom grade.
 - d. Clean components by dry cleaning method using filtered oil-free air or nitrogen until free of visible contamination.
 - e. Immediately package clean, dry components in heat-sealed nylon or polyethylene bags. Seal bags within a second heat-sealed bag.
 - 4. PVDF body components requiring CP-3 level cleaning shall meet the following additional requirements:
 - a. Use virgin PVDF resins free from chemical additives, fillers, property enhancers, and reinforcements such as antioxidants, mold-release agents, colorants, flame retardants, heat stabilizers, lubricants, antistatic agents, pigments, plasticizers, processing aids, and ultraviolet stabilizers.
 - b. Clean PVDF after molding, in an aqueous, nonionic, phosphate-free surface-active agent. Rinse with 18-megohm-cm deionized water and dry with filtered oil-free air or nitrogen.
- E. Level CP-4:
 - 1. Assemble components without lubricants.
 - 2. Seal component in polyethylene bag labeled No Lubricants.
- F. Level CP-5:
 - 1. Chemically passivate/pickle the component to remove surface contaminants and create a relatively inert surface finish.
 - 2. Rinse with RO quality or better water.
 - 3. Confirm passivation quality of wetted surfaces through ferroxyl testing per ASTM A380 using RO quality water to wipe off the test solution from the component interior.
 - 4. Dry with CDA or better quality dry gas.
 - 5. Seal components in polyethylene bag labeled Passivated and Cleaned.
- G. Level CP-6:
 - 1. Electropolish component to remove surface contaminants, surface roughness, and to create a more inert surface finish, with a minimum chrome-oxide to iron-oxide ratio of 1.8 at any wetted point.
 - 2. Rinse with RO quality water, minimum.
 - 3. Visually inspect pipe, tube, or component stub ends interior for contamination, discoloration, or other

rejectable surface anomalies. If unable to correct visual problems by re-cleaning, reject and label the rejected spool.

4. Provide coupons through the electropolish process to certify the process chrome-oxide to iron-oxide ratio, the chrome-to-iron ratio, and the surface roughness.
5. Dry with CDA or better-quality dry gas.
6. Seal components in double polyethylene bags labeled Electropolished and Cleaned.

2.3 TESTING FOR HIGH-PURITY SERVICE

- A. Specific testing criteria apply only to valves and components that specifically reference the criteria in the valve or component data sheet or schedule. The criteria apply to valve and component manufacturing requirements prior to shipment.
- B. Test methods and sampling:
 1. Perform inboard helium leak test by maintaining a 100 percent helium atmosphere in a bag or other enclosure around the component.
 2. Perform outboard helium leak test by maintaining 100 percent helium at 10.5 kg/cm² inside the component.
 3. Perform across-the-seat helium leak testing by maintaining 100 percent helium at atmospheric pressure on the inlet side of the component.
 4. Perform particle testing per ASTM FT394-92.
 5. Testing shall be performed on samples selected per the vendor's sample plan except where noted on data sheets.
- C. Level T-1:
 1. Helium Leak Test:
 - a. Inboard leak rate, less than 10⁻⁹ atm-cc per second.
 - b. Outboard leak rate, less than 10⁻⁶ atm-cc per second.
 2. Across-the-Seat Helium Leak Test: Leak rate, less than 10⁻⁹ atm-cc per second.
 3. Particle Generation - Static Testing: Particle generation per operating cycle (open and closed component), zero counts at 0.3 micron and larger particle size.
- D. Level T-2:
 1. Helium Leak Test:
 - a. Inboard leak rate, less than 10⁻⁷ atm-cc per second.
 - b. Outboard leak rate, less than 10⁻⁴ atm-cc per second.

2. Across-the-Seat Helium Leak Test: Leak rate, less than 10^{-7} atm-cc per second.
 3. Particle Generation - Static Testing: Particle generation per operating cycle (open and closed component), zero counts at 0.3 micron and larger particle size.
- E. Level T-3:
1. Bubble-point tested to 10.5 kg/cm² bubble-tight shutoff.
 2. Test using only DI water and filtered oil-free nitrogen.
- F. Level T-4:
1. Bubble-point tested to 10.5 kg/cm² bubble-tight shutoff.
- G. Shipping Requirements:
1. Guarantee quality and cleanliness upon delivery to jobsite.
 2. Package in wooden crates with voids filled with shock-absorbing materials, shipments of 20 kilograms or greater. Wrap wooden crates in stretch plastic. Durable, reusable plastic cases may be used in lieu of wooden crates.

2.4 INSULATING FLANGES, COUPLINGS, AND UNIONS

- A. Insulating flanges, couplings, or unions shall be provided wherever:
1. Copper and ferrous metal piping are connected.
 2. Cathodically protected steel lines enter buildings
 3. Submerged metallic piping is connected to unsubmerged piping
 4. New galvanized pipe is connected to existing galvanized pipe
 5. Shown on the Drawings.
- B. Insulated joints connecting submerged piping to exposed piping shall be installed above maximum water surface elevation and before the first pipe support not having coated anchor bolts or adhesive-bonded concrete anchors.
- C. Submerged metallic piping shall be isolated from the concrete reinforcement.
- D. Acceptable manufacturers of insulating flanges and unions are Epco and Capitol Insulation Unions. Insulating couplings shall be Dresser STAB-39, R. H. Baker Series 216.

2.5 CORROSION PROTECTION FOR BURIED PIPING, PIPE COATING, AND WRAPPING

- A. Buried ferrous metal piping shall be protected against corrosion using appropriate cathodic protection.
- B. Protective Coatings:
 - 1. Carbon steel piping that will not be insulated shall be painted as required by Contract.
 - 2. Carbon steel pipe installed underground shall be purchased with a shop-applied fully pasticized coal tar enamel coating system in accordance with NAPCA Specification TGF-3. Pipe shall be cleaned in accordance with SSPC-SP6, Commercial Blast Cleaning, and coated and wrapped pipe shall be electrically inspected as specified by AWWA C203. Underground fittings and field welds shall receive an SSPC-SP3 power tool cleaning to remove all loose mill scale, loose rust, loose paint, weld flux, and other loose matter; then primed and tape wrapped in accordance with AWWA Standard C209, Cold-Applied Tape Coatings for Special Sections, Connections, and Fittings for Steel Water Pipelines. Tape shall be a Type I material and shall receive electrical inspection as specified by AWWA C209. Maximum operating fluid temperature - 32 degrees C (90 degrees F).
 - 3. The coating system specified in paragraph 2 above is preferred; however, where approved by the Engineer, the following coating methods for steel pipe may also be used as an alternate.
 - a. Hot-applied, fusion-bonded epoxy coating system in accordance with AWWA C213. Tape coat joints in accordance with AWWA C209. Maximum operating fluid temperature - 40 degrees C (140 degrees F).
 - b. Cold-applied tape coating system in accordance with AWWA C214 Polyken YGIII, or Engineer-approved equal. Tape coat joints in accordance with AWWA C209. Maximum operating fluid temperature - 93 degrees C (200 degrees F).
 - c. Hot applied, 100 percent solids, polyurethane coating system. Maximum operating fluid temperature - 60 degrees C (140 degrees F). Madison Corropipe FSX or Engineer approved equal, Madison Chemical, Inc., St. Louis, Missouri.
- C. Underground cast iron or ductile iron piping buried in areas identified where exceptional exposure to corrosion may occur shall be encased in a polyethylene sleeve as specified in Contract and in accordance with AWWA C105.

2.6 BOLTS, NUTS, AND WASHERS

- A. Bolts, nuts, and washers, and other attachment hardware shall be corrosion resistant, constructed of stainless steel, galvanized or cadmium-plated carbon steel, aluminum-bronze, or similar Engineer-approved material suitable for the service intended or the area of use.
- B. Bolting: Bolts, studs, and nuts shall be contained in marked boxes denoting material and conformance to ASTM A193, ASTM A. Flat washers matching the bolt/nut material shall be used for nonmetallic piping.

Identifier	Component	Material
BC-1	Bolts/studs	Alloy Steel ASTM A193 GR-B7
	Nuts	Heavy hex head ASTM A194 GR-2H
BC-2	Bolts	Carbon steel ASTM A307, GR- B plated
	Nuts	Heavy hex head, plated ASTM 536 GR-B
BC-3	Bolts	Carbon steel, ASTM A307, GR B
	Nuts	Heavy hex head ASTM 536 GR B
BS-1	Bolts/studs	Stainless steel ASTM A193 GR B8A, 304 SS
	Nuts	ASTM A194 GR-8A, 304 SS
BS-2	Bolts/studs	Stainless steel ASTM A193 GR B8M, 316 SS
	Nuts	ASTM A194 GR-8M, 316 SS

2.7 GASKETS

- A. Dimensional Requirements: Gasket dimensions shall conform to the following standards, unless otherwise noted.
1. ASME B16.21: Covers nonmetallic flat gasket dimensions for cast iron and steel pipe flanges of the flat-face, raised-face, large-male and -female, and

- large-tongue-and-groove system through Class 1,500 rating.
2. ASME B16.20: Covers ring-joint gaskets for steel pipe flanges.
 3. ASME B16.5: Covers gaskets other than ring-joint in sizes through 24 inch (600 mm) and ratings through Class 2,500 (17.2 MPa).
 4. ASME B16.47: Covers dimensions for steel flanges in sizes 26 inch (660 mm) through 60 inch (1500 mm) and ratings through Class 900 (6.2 MPa).
 5. AWWA C110/C115: Covers ring and full-face rubber gaskets used with cast and ductile iron flanged pipe and fittings through 54-inch (1350 mm) size.
 6. AWWA C207: Covers dimensions for steel slip-on and plate flanges used for waterworks service in sizes 4 inch (100 mm) through 144 inch (3600 mm).
 7. API 601: Covers double-jacketed corrugated and spiral-wound metallic gaskets.
 8. API 605: Covers dimensions for steel flanges and sizes 26 inch (660 mm) through 60 inch (1500 mm) and ratings through Class 900 (6.2 MPa).
 9. MSS SP-44: Covers dimensions for steel flanges in sizes 12 inch (300 mm) through 36 inch (900 mm) and ratings through Class 900 (6.2 MPa).
 10. Gaskets for large-diameter flanges not covered by ASME B16.21 or B16.5 shall be patterned to match flanges made to AWWA C207, Class E, unless noted otherwise.
- B. The following guidelines establish dimensions for nonmetallic flat gaskets unless otherwise noted:
1. Flat ring gaskets shall be self-centering having an outside diameter equal to the bolt circle diameter less than one bolt diameter.
 2. The gasket inside diameter for steel pipe flanges shall equal the inside diameter of the mating steel pipe. The gasket inside diameter for cast or ductile iron pipe flanges shall equal the nominal pipe size of the mating iron pipe.
 3. Full-face gaskets shall have an outside diameter equal to the outside diameter of the mating flange and an inside diameter as stated above. Holes shall be punched in accordance with the size and number of the mating flange holes.
 4. When a cast/ductile iron flanged pipe is mated to a steel flange, the gasket inside diameter shall equal the nominal pipe size of the mating pipes.
- C. Spiral-wound gaskets shall have a centering-ring outside diameter equal to the bolt circle diameter less one bolt diameter and a gasket inside diameter as specified in

API Standard 601. Gaskets furnished with a solid metal inner ring shall have an inner ring ID that is 3 mm (1/8 inch) greater than the flange bore unless noted otherwise.

- D. Gaskets must be branded with manufacturer and type in order to identify material and rating.
- E. Gaskets Materials: The following groups designate the gasket types and approved manufacturers for each group. There are no alternatives acceptable.

Identifier	Service
G-1	General service.
	Nonasbestos. Durabla 8500. Frenzelit 825. Garlock 3000. Flexitallic SF 2400.
G-2	Select corrosive and nonflammable solvent services.
	Specially compounded Teflon. Durabla 9000. Garlock 3504. Flexitallic Sigma 500.
G-3	High purity water and select chemical services.
	Virgin, un-pigmented PTFE encapsulating either an EPDM or Viton elastomer. Sizes 1/2 inch (25 mm) to 12 inch (300 mm), use 1/8-inch (3 mm) thick Garlock Stress Saver or Asahi AV. Provide cleaned for high-purity service, purged in double bags.
G-4	Steam, condensate high-temperature hot water.
	Flexitallic CG. Leader LG-13. Signum CD.
G-5	Corrosives (typically acids) and selected chemical service 1/8-inch (3 mm) thick with Old ANSI dimensions.
	Viton, 75 Durometer 6.9 MPa minimum tensile strength. Reference preferred material: Thermodyn TD70C.
G-6	EPDM, caustic and other selected services.

Identifier	Service
	1/8-inch (3 mm) thick with Old ANSI dimensions. EPDM (EPT, EP). 60 to 65 Durometer. 6.9 MPa minimum tensile strength.
G-7	Fire-safe.
	Special (PTFE) filled spiral wound. Flexitallic LS Combo Flexcb/PTFE. Signum high purity. Soft seating, fire tested.
G-8	No hub.
	One-piece molded neoprene, ASTM C564 sleeve with sealing beads.
G-9	Full-face, Class 125, Old ANSI, 1/8-inch (3 mm) thick EPT rubber.
G-10	Large-diameter piping for water service. Nitrile (NBR) with synthetic reinforcement. 17.2 MPa tensile strength. 1/8 inch (3 mm) thick.
	Durabla 8500 Garlock Blue-Gaurd 3000. Flexitallic SF 2400.
G-11	AWWA C111 Rubber push-on joint.
G-12	One piece compression gasket, molded neoprene, ASTM C564.

- F. Contractor shall furnish the Engineer with the constants G_b , a , and G_s for each type of gasket. The gasket supplier shall calculate for each gasket size and type the sealing requirements and bolt torque based on the design criteria listed in Contract and the following criteria:
1. Maximum allowable bolt stress is based on 60 percent of yield.
 2. New nuts and bolts: nonmetallic lubricant is used on bolts, nuts and washers.
 3. Include washers for nonmetallic pipe.
 4. These calculations shall be based on the PVRC (Pressure Vessel Research Council) Convenient method.

2.8 THREAD SEALANTS

- A. The following thread sealant groups shall be used as identified in Contract:

Identifier	Thread Sealant
TS-1	1 inch (25 mm) and under - Teflon tape Over 1 inch (25 mm) - Rectorseal T plus 2.
TS-2	Rectorseal T plus 2.
TS-3	Rectorseal 7 (wait 24 hours before pressure).
TS-4	Joint manufacturer standard (push-on joints).
TS-5	Teflon tape.

2.9 SERVICE SADDLES

- A. General: Service saddles may be used only when allowed in the applicable detail piping system specification or specifically approved by Engineer prior to use.
- B. Service Saddles for Ferrous Metal Piping: Pipe service saddles for ferrous metal piping except stainless steel shall be Rockwell Series 313 or 366, Dresser Style 91. Service saddles shall be capable of withstanding 10.5 kg/cm² internal pressure without leakage or overstressing. The run diameter shall be compatible with the outside diameter of the pipe on which the saddle is installed. Taps shall have American National pipe threads. Saddles shall have malleable or ductile iron bodies and galvanized steel straps, steel hex nuts with washers, and neoprene seals. Service saddles shall be of double-strap design.
- C. Service Saddles for Plastic Pipe: Service saddles for CPVC, PVC, and polypropylene pipe shall not be used without written approval from the Engineer.

2.10 AIR VENTS AND LOW-POINT DRAINS

- A. Water line vents and drains required for normal operation are shown on the piping and instrumentation drawings (P&IDs). However, vents and drains at high and low points in the piping that are required for complete venting and draining may not be shown on these Drawings. The Contractor shall add such items during detail piping design and/or piping installation.
- B. Unless otherwise noted in Contract, vent the high point and drain the low point with 20 mm (3/4-inch) block valves on those pipelines 80 mm (2-1/2 inches) and larger, and 12

mm (1/2-inch) block valves on those pipelines 50 mm (2 inches) and smaller; valve types to be as required by Contract. Each drain or vent valve shall be plugged with a pipe plug or nipple and threaded cap.

2.11 FLOW SENSORS

- A. Piping adjacent to flow sensors shall be installed in accordance with the requirements of the manufacturer of the flow sensor and commonly accepted design practices.

2.12 SAFETY CONSIDERATIONS

- A. Piping that must be installed across aisles or other main access walkways shall be covered with a protective checkered plate steel ramp, pitched a maximum of 2 on the vertical to 12 on the horizontal to allow safe wheeled or foot traffic, and painted with a prime coat and two finish coats of safety yellow. Ramp shall be supported from the floor and shall not rest directly on the pipe.
- B. Liquid piping shall not be installed immediately over or within a one meter (3-foot) plan view clearance of any electrical panel, motor starter, or control panel. Where piping must be located within these zones, either install piping inside a PVC conduit or shield the electrical device to prevent direct liquid access to electrical equipment.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Inspect components with special cleaning/packaging requirements upon receipt by the Contractor's QCR. Inspect the components to verify there is no physical damage, caps are in place, bags are not torn, and the necessary documentation is included with the shipment. Rejected material shall be clearly marked, removed from the construction area, temporarily stored onsite in a separate location, removed from the jobsite in a timely manner, and replaced by the manufacturer.

3.2 FABRICATION AND TESTING MEDIA

- A. When specific liquid or gas media is required by individual piping, valve, or component specification sections, the media quality shall comply with the minimum criteria. Specific liquid and gas media are required in other sections for welding, purging, cleaning, testing,

and packaging. Such applications shall use the media quality requirements of this Section for both factory and field activities.

B. Argon (Ar):

1. Filtered to 0.01 micron. Filter housings shall have 316L stainless steel assemblies with electropolished 10-microinch interior finish and an all-Teflon membrane, membrane support, core, end caps, and seals with VCR or extended weld ends.
2. Oxygen impurities less than 100 ppb.
3. Moisture impurities less than 100 ppb.
4. Total hydrocarbon (THC) impurities less than 100 ppb.
5. Particles 0.1 Micron and Larger: less than 30 particles per cubic meter.

C. Nitrogen (N₂):

1. Filtered to 0.01 micron.
2. Oxygen impurities less than 1.5 ppm.
3. Moisture: impurities less than 1 ppm.
4. THC impurities less than 1 ppm.
5. Argon may be used in lieu of nitrogen.

D. Helium (He):

1. Filtered to 0.01 micron.
2. Moisture: less than 1 ppm.

E. Clean Dry Air (CDA):

1. Filtered to 0.1 micron.
2. Minus 100 degree F (-73 degrees C) dew point.
3. Non-lubricated compressor.
4. Argon or nitrogen may be used in lieu of CDA.

F. Deionized (DI) Water:

1. 16 meg ohm-cm minimum resistivity.
2. 100 ppb maximum TOC.
3. Filtered to 0.1 micron.

G. Reverse Osmosis (RO) Water:

1. 0.5-megohm-cm minimum resistivity or RO product quality.
2. Filtered to 0.2 micron.

H. Isopropyl Alcohol (IPA): electronics grade.

3.3 FABRICATION, CLEANING, AND ASSEMBLY WITHIN A CLEANROOM

- A. This criteria for activities within a cleanroom applies to those temporary cleanrooms or cleanroom trailers provided by the Contractor for the purposes of piping and equipment

- system fabrication. This criteria does not apply to activities within the Owner's cleanroom facility.
- B. When a classified cleanroom is required by individual piping, valve, or component specification sections, the cleanroom construction and protocol within the cleanroom shall comply with the minimum criteria. Specific cleanroom ratings are required in other sections for fabrication, cleaning, assembly, testing, and packaging. Such applications shall use the cleanroom quality and protocol requirements of this Section for both factory and field activities.
- C. Class 100 Cleanroom:
1. May be a temporary cleanroom or cleanroom trailer. Shall be a laminar-flow cleanroom in compliance with Federal Standard 209E or current equivalent.
 2. Temperature controlled to 22.5 +/- 1 degrees C (72.5 degrees F plus-or-minus 1.8 degrees F).
 3. Humidity controlled to 42 +/- 3 percent relative humidity.
 4. No grinding, buffing, or other particle-generating activities may occur within the cleanroom.
 5. Provide a clean vestibule in which entering and exiting personnel can gown and disrobe.
 6. Provide a vestibule separate from the personnel vestibule for bringing construction materials in and out of cleanroom.
 7. Personnel protocol to include:
 - a. Cleanroom latex gloves at all times, replaced twice daily.
 - b. Use of lint-free paper for documentation needs.
 - c. Cleanroom body-suit covering all street clothes, replaced daily.
 - d. Cleanroom booties, replaced daily.
 - e. Hair and beard/mustache nets, replaced daily.
 - f. Cleanroom head covering that tucks into the body-suit at the neckline, replaced daily.
- D. Class 10,000 Cleanroom:
1. May be a temporary cleanroom or cleanroom trailer. Shall be a laminar-flow cleanroom in compliance with Federal Standard 209E or current equivalent.
 2. Temperature controlled to 22.5 +/- 1 degrees C (72.5 degrees F plus-or-minus 1.8 degrees F).
 3. Humidity controlled to 42 +/- 3 percent relative humidity.
 4. No grinding, buffing, or other particle-generating activities may occur within the cleanroom.
 5. Personnel protocol to include:

- a. Cleanroom latex gloves at all times, replaced twice daily.
- b. Cleanroom gown, replaced daily.
- c. Cleanroom booties, replaced daily.
- d. Hairnet, replaced daily.

3.4 DRAWINGS

- A. Spool drawings indicate the complete line, showing welded and assembly items, except for insulation shoes or non-stress-relieved lines.
- B. Except for ring-joint flanges, dimensions are to the centerline of pipe and the contact face of flanges. This includes the contact face of male and female tongue-and-groove flanges. Ring-joint flanges are dimensioned to the centerline of pipe and to the extreme face of the flange and not to the contact surface of the groove.
- C. Reducers are concentric unless otherwise noted.

3.5 MATERIAL INSTALLATION

- A. Flanges:
 - 1. Flange bolt holes shall be oriented as follows, unless otherwise indicated on the spool drawings.
 - a. Flange Face Vertical: bolt holes to straddle the vertical centerlines.
 - b. Flange Face Horizontal: bolt holes to straddle plant north-south centerlines.
 - 2. Welds at orifice flanges shall have internal surfaces ground smooth to the pipe wall.
 - 3. Slip-on flanges shall be welded inside and outside. There shall be a distance of approximately 2-3 mm (1/16 to 1/8 inch) between the edge of the fillet weld and the face of the flange. The seal weld shall be applied so that the flange face shall be free of weld spatter and does not require re-facing. Drill hole in flange between welds. Hole to be at 6 o'clock position.
 - 4. Use flat-faced flanges when mating to Class 125 flanges. Use full-face gaskets with flat-faced flanges and ring gaskets with raised-faced flanges.
 - 5. Unless otherwise specified in the data sheets, weld-neck flanges shall be used with butt-weld fittings.
 - 6. The bore of weld-neck flanges shall match the pipe wall thickness.
- B. Seal Welding:
 - 1. When seal welding is required, connections shall be made without using sealing compound or Teflon tape.

2. Do not seal-weld threaded joints that have failed a pressure test, unless all thread compound and Teflon tape have been removed.
 3. Seal welds of threaded connections shall cover all exposed threads.
- C. Bolts:
1. Unless otherwise specified bolt threads shall be coated with Never-Seez (catalog No. NSBT-16) prior to being made up with nuts. Alternate coating shall be used only with prior review of the Engineer.
 2. Bolts shall be tightened in a star sequence. Bolts shall be tightened based on three passes of the bolts. Bolts shall be torqued to levels in the tables provided to Contractor by the gasket vendor being used for that service. Bolt torques shall be made with a calibrated torque wrench. Torque wrench shall be recalibrated at the end of each day.
- D. Piping Expansion and Flexibility:
1. Allow for pipe expansion in mains, runouts, and risers by means of natural flexibility in piping swing joints and by expansion loops where required by the design.
 2. Install with sufficient flexibility to avoid or minimize the use of flexible couplings or expansion joints. Flexible couplings or expansion joints shall be provided for piping connections to equipment where shown. Use of additional flexible couplings or expansion joints, other than those indicated on the piping drawings, requires approval of the Engineer.
 3. Support piping in accordance with the requirements of Contract.
- E. Dimensional Tolerances:
1. These tolerances apply to in-line items and connections for other lines.
 2. General dimensions, such as face-to-face, face-on-end-to-end, face or end-to-center, and center-to-center shall be plus-or-minus 3 mm (1/8 inch).
 3. Inclination of flange face from true in any direction shall be 4 mm per meter (3/64 inch per foot).
 4. Rotation of flange bolt holes shall not exceed 1.6 mm (1/16 inch).
- F. General Considerations:
1. Install instruments and specialty items according to the manufacturer's instructions and with sufficient clearance and access for ease of operation and maintenance.
 2. Use flanges or unions for connections to equipment as required by the piping specifications.

3. Isolation valves for equipment and instruments in a manner which will allow ease of access and removal of the isolated items.
4. Install piping without springing or forcing the pipe in a manner which would set up stresses in the pipe, valves, or connected equipment.
5. Where piping connects to equipment, it shall be supported by a pipe support and not by the equipment.
6. Vibration Compensation Joints:
 - a. Installation of all vibration compensation units shall be per manufacturer's requirements.
 - b. Vibration compensation units shall not be used to make up for misalignment of piping, poor workmanship, or pipe-to-equipment connections.
 - c. Retaining (or control) rods supplied on vibration joints and pump bellows connectors shall have their locking nuts tack welded in place at the position set by the manufacturer. These rods shall not be removed or adjusted from their factory settings. When retaining rods are not installed by the manufacturer, the Contractor shall install the rods with sleeves at the settings specified by the manufacturer of the compensator.
 - d. When shipping retainers are supplied on the compensators by the manufacturer, the retainers shall be left in place until after the compensator is installed and the permanent piping supports are completed and any temporary supports are removed. Shipping retainers are typically tack welded to the flanges. Be certain that retaining rods are not removed.
 - e. Do not use gaskets with compensators which have elastomeric or Teflon flange faces.
 - f. Compensators with internal sleeves shall be installed with the fixed end of the sleeve on the inlet side of the compensator.
 - g. Compensators Installation Sequence:
 - (1) Connect a pipe spool with the same face-to-face dimension as the compensator.
 - (2) Install loose piping flange to the compensator spool.
 - (3) Route piping to the loose flange on the compensator spool.
 - (4) Make the final weld in the piping to the compensator spool. Piping shall be fabricated such that no springing or forcing of the pipe is necessary to make this final weld.
 - (5) Install piping anchors, supports and guides as specified on the Drawings.

- (6) Replace compensator spool with actual compensator. A representative of the compensator manufacturer shall witness removal of the compensator spool and shall take measurements to certify that the axial, torsional, and angular alignments are within the manufacturer's allowable tolerance for an acceptable installation. Certification sheets shall be furnished to the Owner after installation.
 - (7) Remove shipping supports (not the control rods) provided on the compensators
 - h. After the installation of the compensators is complete, re-inspect the units to determine if the compensators are within the manufacturer's allowed offset, compression and extension limits or if any units have been pulled tight against the retaining rods or compressed against the sleeves. The inspection shall be repeated after the piping system has been pressurized. Notify the Engineer of units which have moved to the maximum allowed position or are outside the manufacturer's limits, including offset before attempting to adjust the piping.
 - i. Final: Align motor shafts to pumps (if applicable).
- G. Piping that Penetrates Floors, Ceilings, and Walls:
 - 1. Provide chrome escutcheon finish plates where piping passes through walls, floors, or ceilings in finished areas and cabinets.
 - 2. Major piping penetrations of footings, slabs, floors, walls, and roofs shall be as shown on the Drawings. Verify the size and location of building and structure penetrations prior to pouring concrete or finishing. Sealers shall be in accordance with the requirements of Contract.
 - 3. Wall pipes and pipe sleeves embedded in concrete walls, floors, and slabs shall be embedded as required by the design. Support pipes embedded in concrete walls, floors, and slabs with formwork to prevent contact with the reinforcing steel.
 - 4. Galvanized steel pipe sleeves shall be used on standard wall penetrations. Waterproof pipe sleeves manufactured by Link-Seal that incorporate waterproof membrane seals shall be used for piping penetrating exterior walls, roof, and floor slabs on grade.
 - 5. Galvanized or PVC sleeves shall be used on penetrations of footing walls and grade beams as shown on the Drawings.

6. Fire-Rated or Smoke-Rated Pipe Penetrations: Piping that penetrates fire-rated or smoke-rated walls, floors, or ceilings shall be packed with suitable intumescent fire-proofing according to manufacturer's recommendations.
7. Penetrations of chemical resistant coatings on floors or walls shall be sealed in accordance with requirements of Contract.
8. Install parallel to horizontal and vertical building lines except as indicated in the plan drawings. Maintain uniform slope between bottom-of-pipe elevations defined on the Drawings.

3.6 PIPING FABRICATION

A. General:

1. Employ only workers that have been qualified by training and experience to capably perform the specified activities required to accomplish the work in a satisfactory manner.
2. Designate a quality-assurance inspector who is acceptable to the Owner to inspect the installation, cleaning, and testing of the piping systems. The inspector shall also be responsible for maintaining records for spools and appurtenances for inspection and approval by the Engineer.
3. If the Contractor has developed alternative techniques or intends to apply alternative methods considered equivalent to those indicated herein, a proposal on such techniques or methods shall be submitted in writing to the Engineer for review at least 14 days before intended date of use.
4. If there is a conflict between contract documents, request clarification from the Engineer prior to construction.
5. Deviations from Contract require prior review and written approval by the Engineer.

3.7 PRESSURE TESTING

A. General:

1. The pressure testing requirements apply to piping systems.
2. Perform testing on piping after erection, but before installation of insulation. Furnish necessary equipment and material and make taps in the pipe, as required. The Engineer will review the results of these tests as documented by the Contractor. Test pressures, media, and durations shall be as specified in Contract.

3. Unless otherwise required by the Specifications, shop-fabricated pipe spools are not required to be hydrostatically tested at the shop. Spools shall be field tested after erection. The Contractor shall be responsible for the cost of correction of defects revealed during field hydrostatic tests.
 4. The following piping and equipment shall not be subjected to pressure testing:
 - a. Rotating machinery, such as pumps, turbines, compressors, and chillers.
 - b. Pressure-relieving devices, such as rupture discs and pressure safety relief valves when the relief pressure is within 10 percent of the test pressure.
 - c. Vessels that do not satisfy impact requirements at the piping test temperature.
 - d. Locally mounted pressure-indicating gauges, where the test pressure would exceed their scale range.
- B. Testing Buried Piping:
1. Conduct final acceptance tests on buried piping that is to be pressure tested after the trench has been completely backfilled. If field conditions permit, as determined by the Engineer, partially backfill the trench and leave the joints open for inspection and conduct an initial service leak test. The acceptance test shall not, however, be conducted until backfilling has been completed.
 2. Where any section of pipe is provided with concrete reaction blocking, do not conduct the pressure test until at least 5 days have elapsed after the concrete thrust blocking is installed. If high-early cement is used for the concrete thrust blocking, the time may be cut to 2 days.
 3. Buried piping that is to be pneumatically tested or subjected only to an initial service leak test shall have joints exposed for the acceptance test.
- C. Test Procedures:
1. Calibration records for gauges used for testing shall be submitted to the Engineer.
 2. Two pressure gauges shall be installed for each testing system. Gauges used for testing shall be installed as close as possible to the low point of the piping system.
 3. When starting the filling of lines to be hydrostatically tested, vents and other connections that can serve as vents shall be open during filling so that all air is vented prior to applying test pressure to a system.

4. If the maximum operating conditions of piping attached to a vessel are the same as those of the vessel, the piping and the vessel may be tested together. However, if the vessel has different maximum operating conditions, it shall be isolated and tested separately.
 5. Examination for leakage shall be made at joints and connections. The piping system shall not show visual evidence of weeping or leaking. Visible leakage shall be corrected at the Contractor's sole expense.
 6. If the pressure falls after the pressurizing system is shutoff, the source of pressure loss must be determined and corrected. The system must be able to hold the test pressure for the test duration required by Contract without detectable loss.
 7. Piping designed for vapor or gas which is specified to be hydrostatically tested shall be provided with additional temporary supports, if necessary, to support the weight of the test liquid.
 8. If the ambient air temperature is less than 4 degrees C (40 degrees F) at the time of the pressure testing, the test medium shall be heated as required to achieve the following temperatures when filling is complete:
 - a. For pipe wall thickness 25 mm (1 inch) or less, 21 degrees C (70 degrees F) minimum.
 - b. For pipe wall thickness greater than 25 mm (1 inch), 38 degrees C (100 degrees F) minimum.
- D. Special Requirements for Pneumatic Testing with Any Gaseous Media:
1. The Contractor shall recognize the hazards associated with compressible fluid testing and shall take necessary precautions to protect all personnel. Piping to be tested shall be secured to prevent damage to adjacent piping and equipment in the event of a joint failure. Instruments or devices that may be damaged by the test shall be removed from the piping or suitably isolated prior to applying the test. Prior to starting the test, notify the Engineer.
 2. A preliminary pneumatic test, not to exceed 1.7 kg/cm², shall be applied to the piping system prior to final leak testing as a means of locating major leaks. Examination for leakage, detected by soap bubbles, shall be made at joints and connections. After visible leaks have been corrected, the pressure in the system shall gradually be increased to not more than one-half of the test pressure, after which the pressure shall be increased in steps of approximately 1/10 of the test pressure until the required test pressure has been reached. The pneumatic test

pressure shall be continuously maintained for a minimum duration of 10 minutes, or as specified in Contract and for such additional time as may be necessary to conduct an examination for leakage. The piping system, exclusive of possible localized instances at pump packing, shall not show evidence of leakage. Visible leakage shall be corrected at the Contractor's sole expense.

E. Testing Media Requirements:

1. Clean, fresh, city water shall be used for hydrostatic testing where water is listed as the test media.
2. Other hydrostatic and pneumatic test media, listed as DI, Ar, and N₂, or other abbreviations, shall have minimum quality standards in compliance with this Section.
3. Water used for testing austenitic stainless steel materials shall not contain chlorine above 100 ppm.
4. After hydrostatic testing, all water shall be drained immediately. Care shall be taken not to pull a vacuum during draining; open all vents.

F. Test Repairs:

1. Materials such as gaskets, bolting, etc., damaged during tests and flushing shall be replaced.
2. New gaskets shall be used each time a flanged joint is made up.
3. Welded joints that are defective shall be repaired in accordance with the applicable pipe specifications. Repaired components shall be reexamined by the original method to determine freedom from defects, and repaired joints shall be retested. Costs for such repair shall be the responsibility of the Contractor.

G. Test Records: Records shall be made by the Contractor for each piping installation. These records shall include at a minimum the following items:

1. Date of test.
2. Description and identification of piping tested.
3. Test fluid.
4. Test pressure.
5. Test duration.
6. Remarks to include such items as:
 - a. Leaks (type and location).
 - b. Repairs made on leaks.
7. Signature and date of person witnessing the test.
8. Certification by Contractor and reviewed by the Engineer.

- H. Welded joints that are defective shall be repaired in accordance with the applicable requirements. Repaired components shall be reexamined by the original method to determine freedom from defects, and repaired joints shall be retested. Costs for such repair shall be the responsibility of the Contractor.

3.8 CLEANING OF PIPING SYSTEMS

- A. Following assembly and testing, piping systems not specifically listed as not to be wetted shall be flushed with water to remove any debris and other foreign material. Flushing velocities shall be a minimum of 0.8 meters per second (2.5 feet per second).
 - 1. Where required by Contract, cone strainers shall be inserted in the connections to attached equipment and left there until cleaning has been accomplished to the satisfaction of the Engineer.
 - 2. Lines designated as requiring drying shall be dried immediately after the completion of the flushing. Before forced drying is started, control valves and in-line instruments which may be damaged shall be removed and replaced as necessary with pipe spools. Instrument air of minus 40 degrees C (minus 40 degrees F) dew point or nitrogen shall be used to purge the piping until a minus 29 degrees C (minus 20 degrees F) dew point is reached. Each low-point drain and the end of each branch line shall be blown until the minus 29 degrees C (minus 20 degrees F) dew point is reached. After each section of piping is dried and approved by the Engineer, the in-line instruments shall be replaced.
- B. General Comments:
 - 1. Reference to the method of cleaning required for a given system will be indicated in the service index or design documents.
 - 2. Allow the Owner or the Owner's Representative access to the premises to witness all cleaning and testing operations.
 - 3. Contractor is responsible for all pumps, compressors, etc., to provide the motive force for cleaning and flushing piping systems. Electrical or other fuel requirements for operation of the cleaning equipment shall be provided by Contractor.
 - 4. Permanent plant equipment shall be blanked off, and spool pieces shall be fabricated to provide a pathway around each piece of equipment. Common types of equipment may share the same spool piece and they shall be moved from piece to piece until the

respective supply and return lines are cleaned.
Removal of temporary blanks shall be the responsibility of the Contractor.

5. Boil out heating water boiler using the manufacturer's recommended procedures and chemicals. Contractor shall coordinate this procedure with the manufacturer's startup personnel.
6. Identify and install temporary branches, fittings, vents, drains, etc., required for performance of cleaning shall be by the Contractor.
7. Control valves and other in-line devices which may be damaged by the cleaning method employed shall be removed or bypassed and cleaned separately.
8. Sections of piping which are removed for temporary connections shall be incorporated in the piping circuit being cleaned, or cleaned separately.
9. Return the piping system to its original integrity following the cleaning process.
10. Procedures listed are recommended procedures only. The Contractor may optionally submit proposals which include alternate cleaning procedures. Alternate procedures shall be subject to approval by the Engineer. Regardless of the procedure used, the Contractor is responsible for achieving the required cleanness.
11. Safety Considerations:
 - a. Contractor shall hold a safety and coordination meeting attended by all cleaning personnel to assure the site safety rules are adhered to. These rules typically include posting cleaning schedules prominently and well in advance, confined space entry permit procedures, audible warnings, flashing lights, etc.
 - b. Notify the Owner of special safety precautions that must be taken to protect non-cleaning personnel working in the area of systems being cleaned. Take steps necessary to protect these people from the hazards identified.
 - c. Designate pipe systems utilizing steam blowdown or hot liquid or hot vapor cleaning to accommodate thermal expansion. This shall be done even if the system normally operates at ambient temperature.
12. Environmental Considerations:
 - a. Contractor is required to manage the disposal of leftover and used chemicals incorporated in the cleaning procedures performed. This management shall meet all federal, state, and local regulations and must be in a written form approved by the Owner before any cleaning is started.

- b. Use of chemicals classified by the EPA as non-hazardous is preferred for onsite cleaning. Variance from this must be approved in writing by the Owner.
- C. Cleaning Methods and Procedures: The cleaning methods apply to field-installed piping systems that specifically reference the cleaning methods.
 - 1. Water Flush (F1).
 - a. Sufficient clear water shall be forced through the piping system to achieve 100 percent of normal operating system velocity, if possible. Otherwise, flushing velocity shall be a minimum of 0.9 meters per second (3 feet per second).
 - b. Flushing shall continue until the water discharged to the process sewer is visibly clean.
 - 2. Compressed Air Blow (F2).
 - a. Perform at least three cycles of 3 minutes blowing time each with a blowing duration and velocity as long and as high as achievable with the available air source. The system can be pressurized and then quickly opened to enhance blowing velocity if needed.
 - b. Place a sounding board target approximately one meter (3 feet) from the discharge point. Fine particulate blown from the pipe will strike this board and make an audible tapping noise until all particulate has been purged from the system.
 - c. Continue blow cycles until the discharge is visibly and audibly clean.
 - 3. Dry Air or Nitrogen Blow (F3).
 - a. Perform at least three cycles of 3 minutes blowing time each with a blowing duration and velocity as long and as high as achievable with the available dry air or nitrogen source. The system can be pressurized and then quickly opened to enhance blowing velocity if needed.
 - b. Continue blow cycles until the discharge is visibly clean and free of residual moisture at the downstream end of the blowdown line. Dryness shall be verified by placing a white cloth target at the discharge point during the last blow cycle. The target shall have no noticeable moisture at the end of this cycle. More-rigorous dryness verification shall be performed on noted pipe systems. Such systems shall be blown with instrument-quality air until dry, as indicated by comparison of the discharge air dew point with the inlet air dew point.

- c. This procedure may be specified for use in conjunction with other cleaning processes (e.g., water flush, etc.). In such applications, this procedure is being performed to dry the pipe and shall be done last. The cycling and velocity criteria stated above may be waived in these cases.
- 4. Detergent Degreasing (F4).
 - a. Flush fresh water through the system to remove excess debris and loose scale per the F1 method. Drain system to process sewer.
 - b. Fill system again with detergent solution consisting of liquid degreaser (Dowell Schlumberger Inc. NDT 190 or equal) diluted at a ration of one to one thousand with water.
 - c. Circulate solution through system for a minimum of 1 hour. Where recirculation is not practical, fill system with solution and allow to soak for a minimum of 3 hours.
 - d. Drain system to process sewer and flush with water until discharge is visibly clean and clear of detergent.
- 5. Liquid Phase Degreasing (F5).
 - a. To be performed by a contractor who specializes in this type of work, who has the suitable equipment and is experienced working with the various types of cleaning agents described.
 - b. Flush fresh water through the system to remove excess debris and loose scale. Drain system to process sewer.
 - c. Fill system again with fresh water and begin circulation. Inject a concentrated solution of caustic soda and wetting agents into the system. This concentrated solution shall yield 2 to 3 percent NaOH, 1 to 2 percent tri-sodium phosphate (Na_3PO_4), and 0.025 to 0.1 percent wetting agent (Dowell Schlumberger, Inc. F-57, or approved equal).
 - d. Heat system to 85 ± 3 degrees C (185 degrees F plus-or-minus 5 degrees F) and continue to circulate. Maintain heat and continue process until field tests indicate complete removal of organics such as oil, mill varnish, paint, etc. (4- to 6-hours circulation anticipated). It should be cautioned that this type of cleaning will not remove the lacquer with which some manufacturers coat their products.
 - e. Drain system and flush with water until pH reaches 7.5 to 8.0. Discharge must be disposed of off

site or neutralized prior to draining to process sewer.

6. Liquid Phase Descaling, Derusting, and Passivation (F6).
 - a. To be performed by a contractor who specializes in this type of work, who has the suitable equipment and is experienced working with the various types of cleaning agents described.
 - b. Complete the liquid phase degreasing process.
 - c. Fill system with a 5 percent inhibited hydrochloric acid, 0.025 percent ammonium bifluoride, and 0.1 percent acid inhibitor (Dowell Schlumberger Inc. A-120, or approved equal) (do not use HCl on stainless steel pipe).
 - d. Circulate this solution at 65 +/- 3 degrees C (150 degrees F plus-or-minus 5 degrees F) until field tests indicate acid equilibrium and complete iron pickup (3 to 4 hours circulation anticipated).
 - e. Filter chemical cleaning solutions and flushing water through a 140-micron (100-mesh) full-flow filter while circulating.
 - f. Displace acid solution to waste with warm water until pH of effluent equals pH of influent. As an alternate, drain acid solution using nitrogen to displace. Refill system with warm water and drain to flush. Repeat flush until a discharge pH of 6 to 7 is achieved.
 - g. Inject a non-concentrated passivation solution of 0.25 percent monosodium phosphate, 0.25 percent disodium phosphate, and 0.50 percent sodium nitrite into system and circulate at 68 to 71 degrees C (155 to 160 degrees F) (2 hours anticipated).
 - h. Drain system using nitrogen to displace and check all low points to ensure complete drainage.
 - i. Blow all lines with dry, oil-free nitrogen to remove residual traces of moisture. Dryness shall be verified by placing a white cloth target at the discharge point during the last blow cycle. The target shall have no noticeable moisture at the end of this cycle.
 - j. Maintain a 0.2- to 0.4 kg/cm² dry nitrogen content until system is put into service.
 - k. Discharge must be disposed of off site or neutralized prior to draining to process sewer.
7. Service Flush (F7).
 - a. Follow normal system startup procedures.

- b. Discharge end points of system into tank or process sewer where they can be visually inspected.
 - c. Continue flushing system until discharge is visibly clean.
- 8. General Steam Blow (F9).
 - a. Safety Considerations - The following precautions shall be taken into consideration:
 - (1) The blow nozzle shall be aimed so that personnel and equipment are not endangered by the debris being discharged at high velocity during the blowing period.
 - (2) The entire blowing procedure must be coordinated with the plant to ensure minimum disruption.
 - (3) Personnel working in the area of the steam blow shall be equipped with hearing protection because of the noise created by the high blowing velocities.
 - b. Blow Pipe Restraints: The temporary blow pipe near the exit nozzle shall be properly supported to withstand the thrust forces created by the velocity and pressure of the steam during the blow.
 - c. Blowing Velocity: The generally accepted criteria for blowing is to achieve mass-velocity in the line at least equal to that at design operating conditions.
 - d. Procedure - The procedure for cleaning a steam line involves the following three major sequences:
 - (1) Prewarm-up.
 - (2) Warm-up.
 - (3) Blow.
 - e. Thermal stresses caused by fairly rapid heating and cooling of the piping also help to break loose weld slag and mill scale during the procedure. For this reason the insulation should be left off of the pipe until the blowing procedure is complete. Some insulation may be required for personnel protection at various locations on the pipe. If the pipe is completely insulated, the cooling time will be extended significantly.
 - (1) Prewarm-up prepares the piping system for the actual steam blows.
 - (2) Make sure all in-line instruments, orifice plates, and control valves (disc and seats optional) are removed and temporary spacers, if required, are installed.
 - (3) Make sure drain and warm-up valves are installed at low points in the pipe system in

- order to prevent water hammer during the steam blows. Disconnect steam traps until system has been cleaned.
- (4) Make sure all stops are removed from spring hangers and all supports, guides, and restraints are in place.
- f. The warm-up sequence is used to thermally cycle the pipeline to check for expansion and break scale loose.
- (1) For drain and warm-up valves, remove condensate in order to prevent water hammer during the steam blows.
 - (2) Crack the bypass valve around the main steam valve to allow the line to warm up. It may be necessary to crack the main steam valve to properly warm up the line. Continue to warm up the line until no water comes out of drain and warm-up valves.
 - (3) Monitor the line for movement and expansion. Discontinue if a problem is encountered with pipe movement. Make required modifications before proceeding.
 - (4) The lines shall be thermally cycled (warmed up and cooled down) such that the metal temperature will vary by at least 56 C degrees (100 F degrees).
- g. Blowing cleans the pipeline.
- (1) Warm up the line again using the same procedure as warm-up sequence (line is warm when no water comes out of drain and warm-up valves).
 - (2) After assurance of no water in the pipeline, close all drain and warm-up valves, and slowly open the main steam valve until blowing conditions are reached.
 - (3) The first blow should be a short, light blow to bring the line up to or near operating temperature to check for pipeline expansion and to remove less dense objects with a lower velocity blow.
 - (4) When the blowing conditions are reached, continue blowing for approximately 5 to 10 minutes. Length of blow can be longer or shorter, depending on the quantity of steam available and visual observation of the cleanness of exit steam.
 - (5) Cease the blow by shutting main steam blow valve and bypass. Open drain and warm-up valves and allow pipeline to cool by at least 56 C degrees (100 F degrees). Cooling time

will vary from 1 to 2 hours. If the piping has been insulated, the cooling time may be 3 to 6 hours.

- (6) Repeat for a minimum of two additional free blows. Allow adequate time for cooling between blows. The color of the steam discharged into the atmosphere can be used as an indication of the debris being removed. At first, the steam will appear to be an orange/rust color due to the removal of ferric oxide (Fe_2O_3) from the system.
- (7) Repeat the free-blowing sequence until the steam consistently appears white.

3.9 ACCEPTANCE

- A. The Engineer reserves the right to have any section of the piping system which is suspected to be faulty cut out of the system by the Contractor for inspection and testing. Should the joint prove to be sound, the Owner will reimburse the Contractor on a time-and-material basis as specified in the Contract. Should the joint prove to be faulty, the destructive test shall continue joint by joint until sound joints are found. Costs for replacement of faulty work and/or materials shall be the responsibility of the Contractor.

TABLE 1

LINE CLASS DESIGNATIONS

CONVENTION

1. First Two Letters: Pipe Material Identifier.
2. Third and Fourth Digits: Pipe Data Sheet Unique Identifier.
3. Note: One piping data sheet per line class designation.
4. For double-contained piping systems, use either:
 - a. Inner pipe callout by outer pipe callout, i.e. ST05 x ST04.
 - b. 1/2-inch (12 A) PA04 by 2-inch (50 A) PC03.

Piping Material	Pipe Material Description
AB01	ABS pipe
CC01	CPVC, Schedule 80
CC02	CPVC, Schedule 80, enhanced QA
CI01	Cast iron pipe (no hub)
CI02	Cast iron pipe (hub and - spigot)
CI03	Cast iron high silicon (no hub/ hub and spigot)
CL01	Clay pipe
CN01	Concrete pipe
CS01	A-53F/A-53B blk scrd/bw
CS02	A-53F A-53B blk, sw/bw
CS03	A-53F A-53B, blk scrd/ mech clpg
CS04	A-53F/A-53B galv scrd
CS05	A-106B Sch 80/std wt A-53B scrd/bw
CS06	A-106B Sch 40/std wt A-53B sw/bw
CS07	A-106B Sch 80/40 A-53B sw/bw
CS08	A-106B Sch 80/A-53B scrd/bw
CT01	Copper tubing, soldered (type L AG, type K BG)
CT02	Copper tubing (DWV)
CT03	Copper tubing, cleaned for O2 service, brazed

Piping Material	Pipe Material Description
DI01	Ductile iron pipe
FR01	FRP piping, bell and spigot
FR02	FRP piping, butt-weld
HV01	Galvanized steel tubing
LS01	Polypropylene-lined steel pipe
LS02	Saran-lined steel pipe
LS03	PVDF-lined steel pipe
LS04	TFE-lined steel pipe
PA01	PFA piping
PA02	PFA piping, high purity
PC01	PVC, Schedule 40
PC02	PVC, Schedule 40 (DWV)
PC03	PVC, Clear, Schedule 40
PC04	PVC, Clear, Schedule 40, split fittings
PC05	PVC, Schedule 80
PC06	PVC, Schedule 80, enhanced QA
PC07	PVC, Schedule 80, clean
PE01	Polyethylene piping, socket weld
PE02	Polyethylene piping, butt weld
PF01	PVDF, socket weld
PF02	PVDF, IR butt-weld
PF03	PVDF, high purity, socket weld
PF04	PVDF, high purity, butt weld
PP01	Polypropylene piping, socket weld
PP02	Polypropylene piping, butt weld, SDR11
PP03	Polypropylene piping, butt weld, SDR32
PP04	Polypropylene piping, high purity, butt weld
SS01	304L Stainless steel piping
SS02	316L Stainless steel piping
SS03	304L Stainless steel piping, cleaned for O2 service

Piping Material	Pipe Material Description
SS04	316L Stainless steel piping, cleaned for O2 service
SS05	316L Stainless steel piping, passivated
SS06	316L Stainless steel piping, EP
SS07	316L Stainless steel piping, Class 300
SS08	316L Stainless steel piping, EP, Class 300
ST01	304L Stainless steel tubing
ST02	304L Stainless steel tubing, bright-annealed
ST03	316L Stainless steel tubing
ST04	316L Stainless steel tubing, clean
ST05	316L Stainless steel tubing, high purity, EP
ST06	316L Stainless steel tubing, double melt, high purity, EP
ST07	316L Stainless steel tubing, bright-annealed, high purity
ST10	316L Stainless steel tubing , compression sealed secondary containment
ST11	316L Stainless steel tubing, seal welded secondary containment
TT01	PFA tubing
TT02	PFA tubing, high purity
TT03	PFA tubing, high purity
TT04	Polyethylene tubing
TT05	Polypropylene tubing
TT06	FEP tubing
TT07	PTFE tubing
VJ01	Vacuum-jacketed tubing, coaxial
VJ02	Vacuum-jacketed tubing, tri-axial

- END -